



$$\Pi_{ij} = p \cdot \delta_{ij}$$

$$\Pi_{ij} = \beta_{im} \beta_{jn} \Pi_{mn}$$

$$= \beta_{im} \beta_{jn} p \delta_{mn} =$$

$$= \beta_{im} \beta_{jm} p = p \delta_{ij}$$

$$\underline{\underline{\Pi}} = \underline{\underline{C}} : \underline{\underline{\varepsilon}}$$

$$C_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + \nu (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}),$$

Considerando la simetría de $\underline{\underline{\Pi}}$ y $\underline{\underline{\varepsilon}}$

$$\lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

$$\begin{aligned} \Pi_{ij} &= C_{ijkl} \varepsilon_{kl} = \frac{1}{2} (C_{ijkl} \varepsilon_{kl} + C_{jikl} \varepsilon_{kl}) = \\ &= \frac{1}{2} (C_{ijkl} \varepsilon_{kl} + C_{jiek} \varepsilon_{ek}) \\ &= \frac{1}{2} (C_{ijkl} \varepsilon_{kl} + C_{jiek} \varepsilon_{kl}) = \end{aligned}$$

Por simetría
 $\varepsilon_{ek} = \varepsilon_{ke}$

$$= \frac{1}{2} (C_{ijkl} + C_{jiek}) \varepsilon_{kl}$$

$$C_{ijkl}^* = \frac{1}{2} (C_{ijkl} + C_{jiek}) =$$

$$= \frac{1}{2} [\lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) + \nu (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}) + \lambda \delta_{ij} \delta_{ek} + \mu (\delta_{ie} \delta_{jk} + \delta_{ik} \delta_{je}) + \nu (\delta_{ie} \delta_{jk} - \delta_{ik} \delta_{je})] =$$

$$= \frac{1}{2} \lambda (\delta_{ij} \delta_{kl} + \delta_{ij} \delta_{ek}) + \frac{1}{2} \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} + \delta_{ik} \delta_{jk} + \delta_{il} \delta_{je}) + \frac{1}{2} \nu (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk} + \delta_{il} \delta_{jk} - \delta_{ik} \delta_{je})$$

$$= \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk})$$

$$\Pi_{ij} = C_{ijkl} \varepsilon_{kl} = \Pi_{ji} = C_{jikl} \varepsilon_{kl} \Rightarrow \boxed{C_{ijkl} = C_{jikl}} \Rightarrow$$

$$C_{ijkl} = \frac{1}{2} (C_{ijkl} + C_{jikl}) = \frac{1}{2} (C_{ijkl} + C_{jikl})$$

